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Master's in Applied Econometrics and Forecasting

What Drives House Prices in Luxembourg?

Forecasting and Monetary Policy Evidence

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1 Introduction

Luxembourg has experienced a substantial increase in house prices over the past decades. This development has raised concerns about housing affordability, especially for younger households and first-time buyers. At the same time, recent macroeconomic developments suggest that the housing market may have become more sensitive to changes in interest rates, unemployment, confidence, and overall market activity.

This paper studies the Luxembourg housing market from two complementary perspectives. First, it examines whether macroeconomic fundamentals improve the prediction of house prices compared with a purely univariate time-series benchmark. Second, it investigates the dynamic effect of interest-rate shocks on house price growth.

The contribution of the paper is twofold. First, it evaluates the predictive role of macroeconomic fundamentals using an ARDL/ECM framework and compares it against an ARIMA benchmark. Second, it uses local projections to estimate the dynamic response of house price growth to unexpected interest-rate movements.

2 Data

The analysis is based on quarterly data for Luxembourg from 2007Q1 to 2025Q3. Monthly variables, such as interest rates, unemployment, and consumer confidence, are aggregated to quarterly frequency using quarterly averages. The House Price Index and housing transactions are already available at quarterly frequency.

The main dependent variable is the logarithm of the House Price Index:

$$y_t = \log(HPI_t).$$

The explanatory variables are selected to capture key macroeconomic and housing-market conditions. The interest-rate variable refers to mortgage loan rates for housing loans and is used as a proxy for financing costs in the residential property market. Higher mortgage rates increase borrowing costs for households and may therefore reduce housing demand. The unemployment rate captures labour-market conditions and household income risk, while consumer confidence reflects expectations about the general economic environment. Housing-market activity is measured by the logarithm of the number of housing transactions, denoted by $\log(acts_t)$.

In addition, a dummy variable is included from 2015Q1 onward. This dummy captures the structural break associated with the 2015 tax reform in Luxembourg, which affected the housing market through changes in the fiscal treatment of real estate transactions and investment incentives. The dummy is defined as:

$$D_{2015,t} = \begin{cases} 1, & \text{if } t \geq 2015Q1, \\ 0, & \text{otherwise.} \end{cases}$$

Figure 1 shows the evolution of the logarithmic House Price Index. The series displays a strong upward trend over most of the sample period, indicating persistent house price growth in Luxembourg. However, the pattern also suggests a slowdown and partial correction toward the end of the sample, consistent with the recent weakening of housing-market dynamics.

Figure 2 presents the main macroeconomic explanatory variables. Interest rates decline during much of the pre-2022 period but increase sharply afterwards, reflecting tighter financing conditions. The unemployment rate fluctuates over the sample and captures changes in labour-market conditions. Consumer confidence shows strong cyclical variation, with pronounced downturns during periods of economic uncertainty. Housing transactions vary considerably over time and decline toward the end of the sample, indicating a slowdown in market activity.

In the context of the local projection framework, additional macro-financial variables are incorporated to construct the interest rate shock. Specifically, Euribor and the Harmonised Index of Consumer Prices (HICP) are included to identify unexpected changes in mortgage interest rates. Euribor captures short-term interbank funding conditions and is closely linked to the pricing of mortgage loans, while HICP reflects inflation dynamics and the broader macroeconomic environment. These variables are not directly included as baseline regressors in all empirical specifications but are used to isolate the unanticipated component of mortgage interest rates through residualisation.

Table 1: Variable overview

Variable	Definition	Transformation
$\log(HPI)$	House Price Index	Logarithm
i	Mortgage interest rate	Quarterly average
u	Unemployment rate	Quarterly average
cc	Consumer confidence	Quarterly average
$\log(actes)$	Housing transactions	Logarithm
$Euribor$	Interbank interest rate	Quarterly average
$HICP$	Inflation index	Quarterly average
D_{2015}	Structural dummy	Equals 1 from 2015Q1 onward

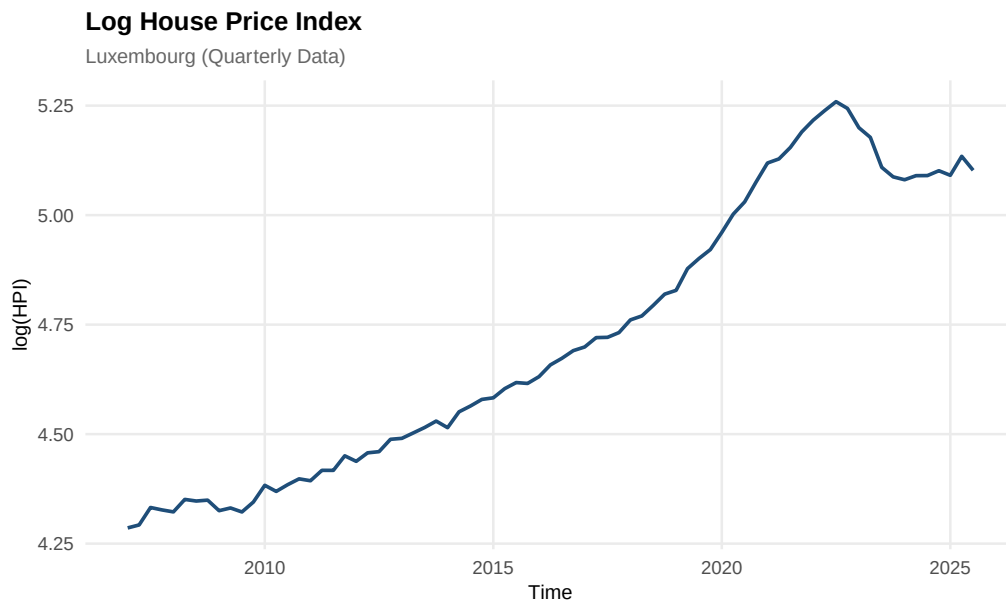


Figure 1: Evolution of the logarithmic House Price Index in Luxembourg.

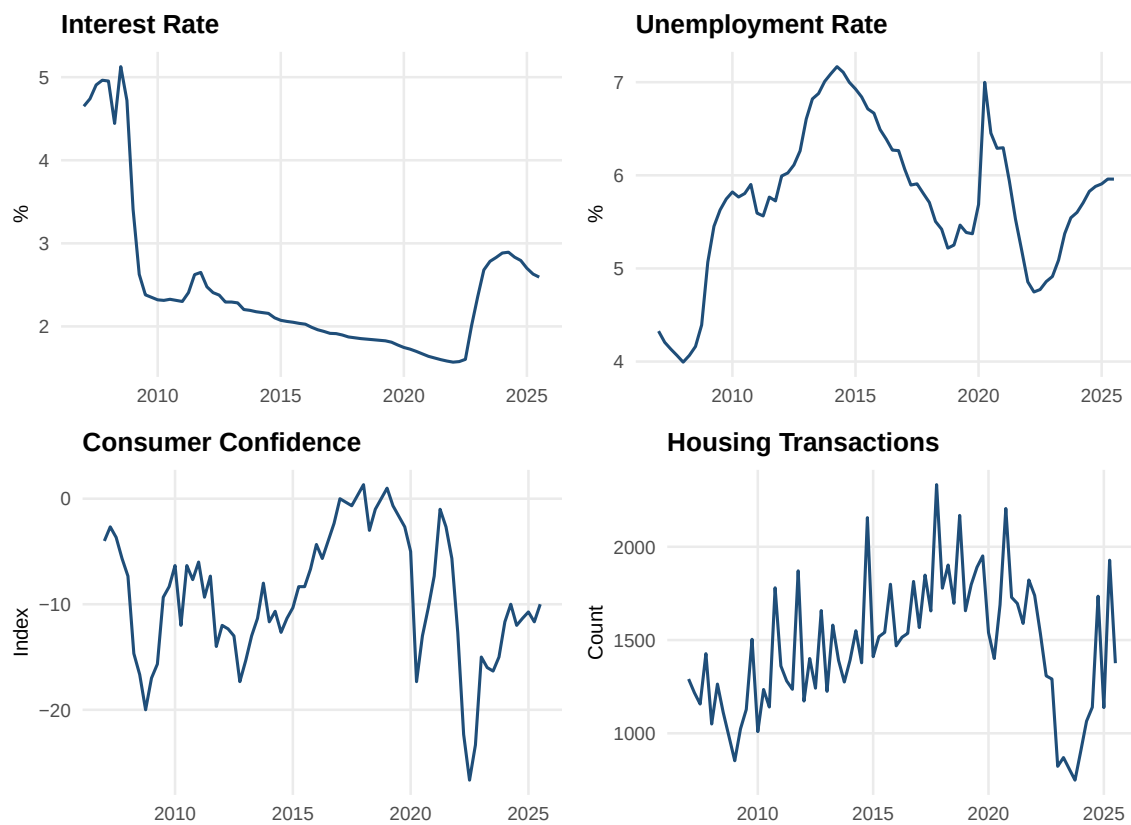


Figure 2: Macroeconomic time series used in the empirical analysis: interest rate, unemployment rate, consumer confidence, and housing transactions.

3 Methodology

3.1 Stationarity Analysis

Before estimating the econometric models, the time-series properties of all variables are examined to determine their order of integration. This step is crucial, as the validity of the ARDL framework depends on the absence of variables integrated of order two, i.e., $I(2)$ processes (Pesaran et al., 2001).

To assess stationarity, two complementary statistical tests are employed: the Augmented Dickey-Fuller (ADF) test (Dickey and Fuller, 1979) and the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test (Kwiatkowski et al., 1992). These tests are used jointly, as they rely on opposite null hypotheses and therefore provide a more robust classification of the integration properties.

The ADF test evaluates the null hypothesis that a given time series contains a unit root, implying non-stationarity. Rejection of the null suggests that the series is stationary. In contrast, the KPSS test is based on the null hypothesis of stationarity, either around a constant (level stationarity) or around a deterministic trend (trend stationarity). Rejection of the KPSS null indicates non-stationarity.

Both tests are conducted under alternative specifications, including a model with a constant (drift) and, where appropriate, a model with both constant and deterministic trend. The choice of specification is guided by visual inspection of the time series and economic considerations.

If a variable is found to be non-stationary in levels but stationary after first differencing, it is classified as integrated of order one, $I(1)$. Variables that are stationary in levels are classified as $I(0)$. Only variables that satisfy these conditions are retained for further analysis, ensuring that the dataset contains a mixture of $I(0)$ and $I(1)$ variables but no $I(2)$ processes.

This classification provides the necessary foundation for the subsequent application of the ARDL bounds testing approach, which explicitly allows for regressors with different orders of integration within this range (Pesaran et al., 2001).

3.2 Forecasting Framework

To evaluate the predictive performance of alternative time-series models, a comparison between an Autoregressive Distributed Lag (ADL) model and a univariate ARIMA model is conducted. This comparison allows us to assess whether incorporating macroeconomic fundamentals improves forecasting accuracy relative to a purely data-driven time-series approach.

3.2.1 ADL Model Specification

To model the relationship between house prices and macroeconomic fundamentals, an Autoregressive Distributed Lag (ADL) framework is employed. The ADL model allows for a flexible lag structure, capturing both the persistence in the dependent variable and the dynamic effects of explanatory variables over time.

The general $ADL(p, q)$ specification can be written as:

$$\log(HPI_t) = \alpha + \sum_{i=1}^p \beta_i \log(HPI_{t-i}) + \sum_{j=0}^q \gamma_j X_{t-j} + \varepsilon_t,$$

where X_t denotes a vector of explanatory variables, including the mortgage interest rate, unemployment rate, consumer confidence, housing transactions, and the structural dummy.

The ADL framework is particularly suitable in this context, as it allows for the inclusion of variables that are integrated of order $I(0)$ and $I(1)$, provided that none of the variables is integrated of order two (Pesaran et al., 2001). This flexibility makes the model well-suited for macroeconomic time series, which often exhibit mixed integration properties.

Lag lengths for both the dependent and explanatory variables are selected using the Bayesian Information Criterion (BIC). The BIC provides a consistent model selection criterion by penalising model complexity more strongly than alternative criteria such as the Akaike Information Criterion, thereby favouring more parsimonious specifications (Schwarz, 1978). This is particularly important given the relatively small sample size.

The inclusion of lagged values of the dependent variable captures the strong persistence typically observed in house prices, while the distributed lag structure allows macroeconomic variables to affect house prices both contemporaneously and with delays.

3.2.2 Cointegration Testing (Bounds Test)

To determine whether a long-run equilibrium relationship exists between house prices and the macroeconomic variables, the ARDL bounds testing approach proposed by Pesaran et al. (2001) is applied.

The bounds test evaluates the null hypothesis of no level relationship (no cointegration) against the alternative of a long-run relationship. This is implemented by testing the joint significance of lagged level variables in an unrestricted error correction representation of the ARDL model.

The test is conducted under the specification with an unrestricted intercept and no deterministic trend (case 3), which is appropriate given the characteristics of the data. The resulting F-statistic is compared against critical value bounds that depend on whether the regressors are purely $I(0)$, purely $I(1)$, or a mixture of both.

If the F-statistic exceeds the upper critical bound, the null hypothesis of no cointegration is rejected, indicating the presence of a long-run equilibrium relationship. If it falls below the lower bound, the null cannot be rejected. Values within the bounds lead to inconclusive results.

The bounds testing approach is particularly advantageous in this setting, as it remains valid in the presence of regressors with mixed integration orders and does not require pre-testing for a common integration order beyond excluding $I(2)$ variables.

3.2.3 Error Correction Model (ECM)

Following the detection of a long-run relationship using the bounds testing approach, the ARDL model can be reparameterised into an Error Correction Model (ECM). This representation allows for a clear distinction between short-run dynamics and long-run equilibrium adjustment.

The general ECM specification can be written as:

$$\Delta \log(HPI_t) = \alpha + \sum_j \theta_j \Delta X_{t-j} + \lambda ECT_{t-1} + \varepsilon_t,$$

where Δ denotes the first-difference operator, X_t represents the vector of explanatory variables, and ECT_{t-1} is the error correction term derived from the estimated long-run relationship.

The coefficient λ measures the speed at which deviations from the long-run equilibrium are corrected. A negative and statistically significant value of λ indicates that the system converges back to equilibrium after a shock (Engle and Granger, 1987).

The ECM framework is particularly useful in this context, as it allows short-run fluctuations in macroeconomic variables to influence house price dynamics, while simultaneously accounting for long-run equilibrium forces. This dual structure provides a more comprehensive representation of housing market dynamics compared to purely static or purely differenced models.

The ECM is estimated conditional on the presence of cointegration, ensuring that the long-run relationship identified in the ARDL model is properly incorporated into the short-run specification (Pesaran et al., 2001).

3.2.4 ARIMA Benchmark

As a benchmark model, a univariate Autoregressive Integrated Moving Average (ARIMA) model is estimated for the logarithm of the House Price Index. The ARIMA framework captures the internal time-series dynamics of house prices without incorporating macroeconomic fundamentals.

The general ARIMA(p, d, q) model is defined as:

$$\phi(L)(1 - L)^d y_t = \theta(L)\varepsilon_t,$$

where L is the lag operator, d denotes the order of differencing, $\phi(L)$ and $\theta(L)$ are lag polynomials representing the autoregressive and moving average components, respectively (Box and Jenkins, 1970).

The order of differencing is determined based on the stationarity analysis, ensuring that the series is transformed into a stationary process before estimation. Candidate ARIMA specifications are then selected using information criteria, with a preference for the Bayesian Information Criterion (BIC) to ensure parsimony.

The ARIMA model serves as a purely data-driven benchmark, relying exclusively on past values and shocks of the dependent variable. By comparing its forecasting performance with the ARDL/ECM framework, it is possible to assess whether macroeconomic fundamentals provide additional predictive power beyond autoregressive dynamics.

This comparison is central to the forecasting analysis, as it directly evaluates the contribution of economic variables to the prediction of house price movements.

3.2.5 Forecast Evaluation

The forecasting performance of the competing models is evaluated using a pseudo-out-of-sample framework. The sample is divided into an initial training period and a subsequent evaluation period. Forecasts are generated recursively using an expanding window approach. At each forecast origin, the models are re-estimated and one-step-ahead forecasts are produced.

Forecast accuracy is assessed using two standard loss functions: the Root Mean Squared Error (RMSE) and the Mean Absolute Error (MAE). The RMSE is defined as:

$$RMSE = \sqrt{\frac{1}{T} \sum_{t=1}^T (y_t - \hat{y}_t)^2},$$

while the MAE is given by:

$$MAE = \frac{1}{T} \sum_{t=1}^T |y_t - \hat{y}_t|.$$

The RMSE places greater weight on larger forecast errors due to the squaring of deviations, whereas the MAE provides a linear measure of average forecast accuracy. The use of both metrics allows for a more comprehensive evaluation of predictive performance (Hyndman and Koehler, 2006).

To formally compare the predictive accuracy of competing models, the Diebold-Mariano (DM) test is applied (Diebold and Mariano, 1995). The DM test evaluates the null hy-

pothesis that two forecasting models have equal predictive accuracy against the alternative that one model performs better.

The test is based on the difference in forecast loss between two models. Let $e_{1,t}$ and $e_{2,t}$ denote the forecast errors from two competing models. The loss differential is defined as:

$$d_t = L(e_{1,t}) - L(e_{2,t}),$$

where $L(\cdot)$ is a loss function, such as squared or absolute errors. Under the null hypothesis of equal predictive accuracy, the expected value of d_t is zero.

The DM statistic is constructed as a standardized mean of the loss differential series and follows an asymptotic normal distribution under the null. Rejection of the null hypothesis indicates a statistically significant difference in forecast performance between the models.

This framework allows for both an economic and statistical comparison of forecasting models, providing a rigorous basis for evaluating whether the inclusion of macroeconomic fundamentals improves predictive accuracy relative to a univariate benchmark.

3.3 Local Projections Framework

To analyse the dynamic response of the Luxembourg Housing Price Index (HPI) to unexpected changes in mortgage interest rates, the local projection (LP) methodology introduced by Jordà (2005) is employed. Compared to vector autoregressive (VAR) models, LPs offer greater robustness to model misspecification and are particularly suitable in small samples, as they do not impose dynamic restrictions across horizons.

Shock Identification

Unexpected innovations in mortgage interest rates are constructed via a residual-based approach. Specifically, the following auxiliary regression is estimated:

$$i_t = \alpha + \rho i_{t-1} + \delta_1 Euribor_{t-1} + \delta_2 HICP_{t-1} + \delta_3 u_{t-1} + \varepsilon_t, \quad (1)$$

where i_t denotes the mortgage interest rate, $Euribor_{t-1}$ captures interbank funding conditions, $HICP_{t-1}$ reflects inflation dynamics, and u_{t-1} denotes the unemployment rate. The residual $\hat{\varepsilon}_t$ is interpreted as an unexpected mortgage rate innovation, conditional on past macro-financial information.

This approach follows the reduced-form identification strategy commonly used in empirical macro-finance (e.g., Stock and Watson (2018)), where shocks are defined as innovations orthogonal to an information set available at time $t - 1$. It should be noted that

these shocks are not structural in the sense of fully identified monetary policy shocks, but rather represent conditional innovations.

Local Projection Specification

Impulse responses are estimated using the local projection (LP) methodology of Jordà (2005). For each horizon $h = 0, 1, \dots, H$, the following regression is estimated:

$$y_{t+h} = \beta_h \cdot shock_t + \sum_{j=1}^{p+h} \gamma_{j,h} y_{t-j} + \delta' Z_{t-1} + \xi_{t+h}, \quad (2)$$

where y_t denotes the logarithm of the housing price index, $shock_t$ is the estimated mortgage interest rate innovation, and Z_{t-1} is a vector of lagged macro-financial control variables including mortgage interest rates, Euribor, inflation, and unemployment.

Importantly, the control variables correspond to the same information set used in the construction of the shock. This ensures consistency between the shock identification stage and the local projection regressions, thereby supporting the sequential exogeneity assumption required for consistent estimation of β_h .

Lag Augmentation and Persistence

Given the high persistence and potential non-stationarity of the variables, lag augmentation is applied following ?. Specifically, the number of lags of the dependent variable is allowed to increase with the forecast horizon, ensuring that the regression adequately approximates the underlying data-generating process.

This approach guarantees that the estimator of β_h is asymptotically normal even when the dependent variable exhibits unit root or near-unit-root behaviour. As a result, inference can be conducted directly in levels without requiring pre-differencing or cointegration adjustments.

Importantly, lag augmentation addresses issues related to persistence and inference, but does not by itself ensure identification. Therefore, the inclusion of macro-financial controls remains necessary to support the conditional exogeneity interpretation of the estimated shock.

The baseline lag length p is set to a parsimonious value based on residual diagnostics of an autoregressive model, ensuring the absence of residual autocorrelation according to Ljung–Box tests and inspection of autocorrelation functions.

Inference

Following ?, lag augmentation ensures that the estimator of the impulse response coefficients is asymptotically normal even in the presence of highly persistent or non-stationary dynamics.

Accordingly, heteroskedasticity-robust standard errors (HC1) are used for inference (White, 1980; MacKinnon and White, 1985). While lag augmentation mitigates issues arising from serial correlation induced by overlapping horizons, robust standard errors provide additional protection against heteroskedasticity and potential remaining misspecification.

3.4 Diagnostic Tests

To ensure the validity of the estimated models, a range of diagnostic tests is conducted. These tests assess whether the underlying assumptions of the regression models are satisfied, including the absence of serial correlation, homoskedasticity, normality of residuals, and correct functional form.

First, serial correlation in the residuals is examined using the Breusch–Godfrey test (Breusch, 1978; Godfrey, 1978). This test evaluates the null hypothesis of no autocorrelation up to a specified lag order. Detecting serial correlation is particularly important in time series models, as it may indicate model misspecification or omitted dynamics.

Second, heteroskedasticity is tested using the Breusch–Pagan test (Breusch and Pagan, 1979). The null hypothesis assumes constant variance of the error term. Rejection of the null suggests that the variance of the residuals depends on the explanatory variables, which can lead to inefficient estimates and invalid standard errors if not addressed.

Third, the normality of residuals is assessed using the Jarque–Bera test (Jarque and Bera, 1987). This test is based on the skewness and kurtosis of the residual distribution. While normality is not strictly required for consistency of the estimators, it is important for valid inference in small samples.

In addition, the Ljung–Box test is applied to further examine residual autocorrelation (Ljung and Box, 1978). This test evaluates whether groups of autocorrelations are jointly equal to zero, providing an additional check for remaining serial dependence.

Together, these diagnostic tests provide a comprehensive framework for validating the econometric models and ensuring the reliability of the empirical results.

4 Empirical Results

4.1 Stationarity Results

Table 2 reports the results of the Augmented Dickey-Fuller (ADF) and KPSS tests for all variables in levels and, where necessary, in first differences.

Table 2: Unit Root and Stationarity Tests

Variable	ADF (Level)	KPSS (Level)	ADF (Diff)	KPSS (Diff)
$\log(HPI)$	-1.101	0.20**	-3.291*	0.17
Interest Rate	-2.377	0.33**	-5.407***	0.32
Unemployment Rate	-2.354	0.33	—	—
Consumer Confidence	-3.043*	0.19	—	—
$\log(actes)$	-2.228	0.35	-7.138***	0.06

Notes: ADF reports test statistics. KPSS reports test statistics. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively. For KPSS, significance implies rejection of stationarity.

The results indicate that the logarithm of the House Price Index and the interest rate are non-stationary in levels. For both variables, the ADF test fails to reject the unit-root null hypothesis, while the KPSS test rejects the null of trend stationarity. However, after first differencing, both variables become stationary, as indicated by strongly significant ADF statistics and non-rejection of stationarity in the KPSS test. These variables are therefore classified as integrated of order one, $I(1)$.

For housing transactions, the evidence is also consistent with an $I(1)$ process. While the ADF test in levels does not reject the unit-root null and the KPSS test rejects stationarity under the trend specification, the first-difference series is clearly stationary according to both tests.

In contrast, consumer confidence is found to be stationary in levels. The ADF test rejects the unit-root null, and the KPSS test does not reject level stationarity. The unemployment rate is also treated as level-stationary, as the KPSS test supports stationarity, despite the ADF test failing to reject the unit-root null. This result is not uncommon given the low power of unit-root tests in small samples.

Overall, the dataset contains a mixture of $I(0)$ and $I(1)$ variables, with no evidence of second-order integration. This satisfies the key requirement for the application of the ARDL bounds testing framework.

4.2 Forecasting Framework

4.2.1 ARDL Results

Table 3 reports the preferred ARDL specification and a robustness model including lagged housing transactions. The preferred model excludes housing transactions from the baseline specification to reduce potential simultaneity between prices and market activity. The mortgage interest rate is included only in lagged form, reflecting the delayed transmission of financing conditions to the housing market.

Table 3: ARDL Estimation Results

	Main Model	Robustness Model
$L(\log(HPI), 1)$	1.050*** (0.117)	0.950*** (0.139)
$L(\log(HPI), 2)$	-0.060 (0.116)	0.051 (0.140)
i_{t-1}	-0.006 (0.004)	-0.001 (0.004)
Unemployment rate	0.002 (0.004)	0.003 (0.004)
Consumer confidence	-0.001* (0.001)	-0.001 (0.001)
$L(\text{Consumer confidence}, 1)$	0.002*** (0.001)	0.002** (0.001)
$L(\log(actes), 1)$	— —	0.006 (0.014)
$L(\log(actes), 2)$	— —	0.029** (0.012)
Dummy 2015	-0.001 (0.010)	-0.007 (0.010)
AIC	-362.6	-366.0
BIC	-342.0	-340.8
Adj. R^2	0.996	0.997
Residual Std. Error	0.0189	0.0183
Observations	73	73

Notes: Standard errors are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. The main model excludes housing transactions due to potential simultaneity with house prices. The robustness model includes lagged housing transactions.

The results show that Luxembourg house prices are highly persistent. In the main model, the first lag of $\log(HPI)$ is positive and highly significant, while the second lag is negative but statistically insignificant. Taken together, the autoregressive coefficients imply very persistent house price dynamics. This confirms that current house prices are mainly explained by their own past values.

The lagged mortgage interest rate has the expected negative sign in both specifications, but it is not statistically significant. This suggests that, within the ARDL framework, short-run house price dynamics are not strongly explained by lagged financing costs once house price persistence and other macroeconomic controls are included.

The unemployment rate is not statistically significant in either specification. Similarly, the 2015 dummy is insignificant, suggesting that the tax reform period does not generate a separate level effect once the dynamic structure of the model is controlled for.

Consumer confidence shows a more meaningful pattern. In the main model, the contemporaneous coefficient is negative and weakly significant, while the first lag is positive and highly significant. This suggests that confidence effects may operate with a delay: weaker confidence may coincide with short-run market pressure, while improved confidence in the previous quarter is associated with higher current house prices.

The robustness model adds lagged housing transactions. The first lag of transactions is not statistically significant, while the second lag is positive and significant. This indicates that housing-market activity may contain predictive information for house prices with a delay. However, since transactions and prices are likely to be jointly determined, these results are interpreted as robustness evidence rather than as the baseline causal relationship.

4.2.2 Cointegration Results

Table 4 reports the ARDL bounds test results for the preferred main model and the robustness specification including lagged housing transactions.

Table 4: ARDL Bounds Test for Cointegration

Model	F-statistic	p-value	Interpretation
Main Model	3.3	0.10	Weak / inconclusive evidence
Robustness Model	3.7	0.06	Borderline evidence

Notes: The bounds test is conducted under case 3, allowing for an unrestricted intercept and no deterministic trend. The null hypothesis is no cointegration.

The bounds test results provide only weak evidence for a long-run relationship between house prices and the macroeconomic fundamentals. For the preferred main model, the F-statistic is 3.3 with a p-value of 0.10, which does not provide strong evidence against the null hypothesis of no cointegration. The robustness model including lagged housing transactions yields a slightly higher F-statistic of 3.7 with a p-value of 0.06, suggesting borderline evidence of cointegration.

Compared with earlier specifications, which produced much stronger evidence of cointegration, the results indicate that the long-run relationship is sensitive to model specification. In particular, once additional house price dynamics are included, the interest rate is restricted to enter only in lagged form, and housing transactions are treated more cautiously, the evidence for cointegration becomes substantially weaker.

Therefore, the existence of a stable long-run equilibrium relationship cannot be confirmed with strong confidence. The long-run coefficients are consequently interpreted with caution and are not treated as the main empirical evidence.

4.2.3 Error Correction Model

Although the bounds test provides only weak and borderline evidence of cointegration, an error correction representation is estimated to examine whether the implied adjustment dynamics are consistent with a long-run equilibrium relationship. The results are reported in Table 5.

Table 5: Error Correction Model Results

	Main Model	Robustness Model
Intercept	0.070*** (0.015)	-0.253*** (0.053)
$\Delta L(\log(HPI), 1)$	0.060 (0.112)	-0.051 (0.125)
Δ Consumer confidence	-0.001** (0.001)	-0.001 (0.001)
$\Delta L(\log(actes), 1)$	— —	0.006 (0.011)
ECT_{t-1}	-0.009*** (0.002)	0.001*** (0.000)
Adjusted R^2	0.273	0.334
Residual Std. Error	0.0184	0.0176

Notes: Standard errors are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

In the main model, the error correction term is negative and statistically significant. This sign is consistent with adjustment toward a long-run equilibrium: deviations from the implied equilibrium relationship are gradually corrected over time. The estimated coefficient of approximately -0.009 suggests that around 0.9% of disequilibrium is corrected per quarter. This adjustment speed is very slow, which is consistent with the high persistence observed in Luxembourg house prices.

The short-run change in consumer confidence is negative and statistically significant in the main model. This suggests that deteriorations in consumer sentiment are associated with lower short-run house price growth. However, the lagged change in house prices is not statistically significant, indicating that short-run adjustment dynamics remain limited once the error correction term is included.

The robustness model including lagged housing transactions produces a positive and statistically significant error correction term. This is problematic from an error-correction perspective, because a positive adjustment coefficient implies divergence rather than convergence back to equilibrium. Therefore, the ECM results for the robustness model should not be interpreted as evidence of a stable long-run adjustment mechanism.

Overall, the ECM results support a cautious interpretation. While the main model produces an error correction term with the expected negative sign, the bounds test evidence is only weak, and the robustness model does not display economically meaningful adjustment. Hence, the evidence for a stable long-run equilibrium relationship is not robust across specifications.

4.2.4 ARIMA Benchmark

As a univariate benchmark model, an ARIMA specification is estimated for the logarithm of the House Price Index. Based on the stationarity analysis, $\log(HPI)$ is treated as an integrated series of order one. Therefore, the ARIMA model is estimated with one difference, $d = 1$.

To guide the ARIMA identification, the ACF and PACF of the differenced log house price series are inspected. Figure 3 shows that the differenced series displays only limited remaining autocorrelation, with some short-run dependence at the first few lags. This supports the use of a low-order autoregressive specification.

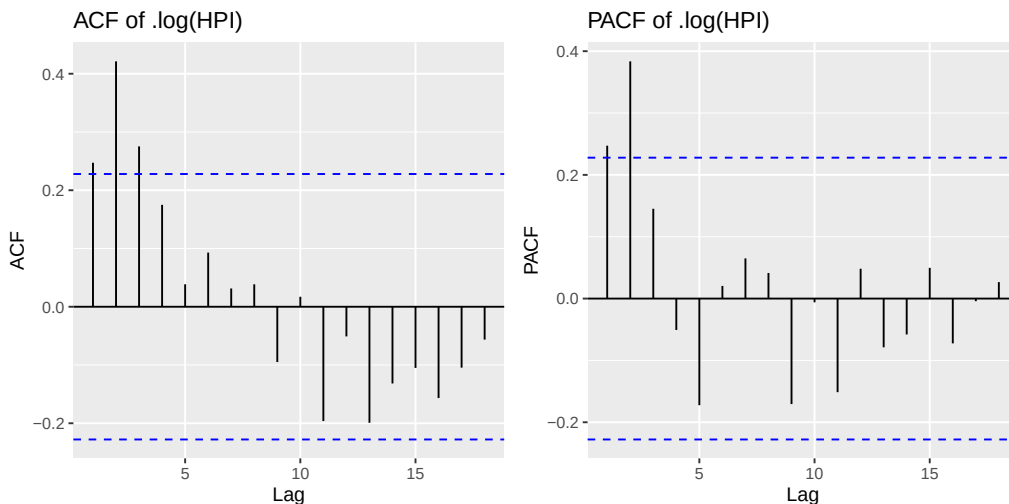


Figure 3: ACF and PACF of the differenced log House Price Index.

The final benchmark model is selected using the Bayesian Information Criterion (BIC). The selected model is an ARIMA(2,1,0), implying that the first difference of $\log(HPI)$ is modelled as an autoregressive process of order two:

$$\Delta \log(HPI_t) = \phi_1 \Delta \log(HPI_{t-1}) + \phi_2 \Delta \log(HPI_{t-2}) + \varepsilon_t.$$

Table 6 reports the estimation results.

Table 6: ARIMA(2,1,0) estimation results

Parameter	Estimate	Std. Error
AR(1)	0.199	0.106
AR(2)	0.484	0.110
AIC	-367.5	
BIC	-360.6	
RMSE	0.0192	
MAE	0.0150	

Both autoregressive coefficients are positive, suggesting that house price growth exhibits short-run persistence. The second autoregressive coefficient is particularly relevant, indicating that changes in house prices depend not only on the previous quarter but also on dynamics two quarters earlier.

Overall, the ARIMA(2,1,0) model provides a parsimonious benchmark that captures the univariate dynamics of house prices without relying on macroeconomic explanatory variables.

4.2.5 Forecast Evaluation

To evaluate the predictive performance of the competing models, a pseudo out-of-sample forecasting exercise was conducted using an expanding window approach. Starting from an initial estimation sample of 60 observations, both the ARIMA benchmark and the ARDL model were recursively re-estimated at each forecast origin. One-step-ahead forecasts of the logarithm of the House Price Index were then generated.

For the ARIMA model, forecasts are purely based on the univariate time series dynamics of $\log(HPI)$. In contrast, the ARDL model incorporates macroeconomic fundamentals. Since contemporaneous values of some explanatory variables are used, the ARDL forecasts should be interpreted as *conditional forecasts*, i.e. forecasts conditional on the observed macroeconomic information available at the forecast date.

Forecast accuracy is evaluated using the Root Mean Squared Error (RMSE) and the Mean Absolute Error (MAE), as reported in Table 7.

Table 7: Pseudo out-of-sample forecast performance

Model	RMSE	MAE
ARIMA	0.0360	0.0286
ARDL (conditional)	0.0345	0.0267

The ARDL model achieves slightly lower forecast errors compared to the ARIMA benchmark, indicating a modest improvement in predictive accuracy when macroeconomic variables are included.

To formally assess whether this improvement is statistically significant, a Diebold-Mariano test is conducted. The test statistic ($DM = 0.8$) is not statistically significant, with a p -value of 0.4. Hence, the null hypothesis of equal predictive accuracy between the two models cannot be rejected.

Figure 4 illustrates the pseudo out-of-sample forecasts. Both models closely track the actual evolution of house prices, with only minor differences between the ARIMA and ARDL predictions.

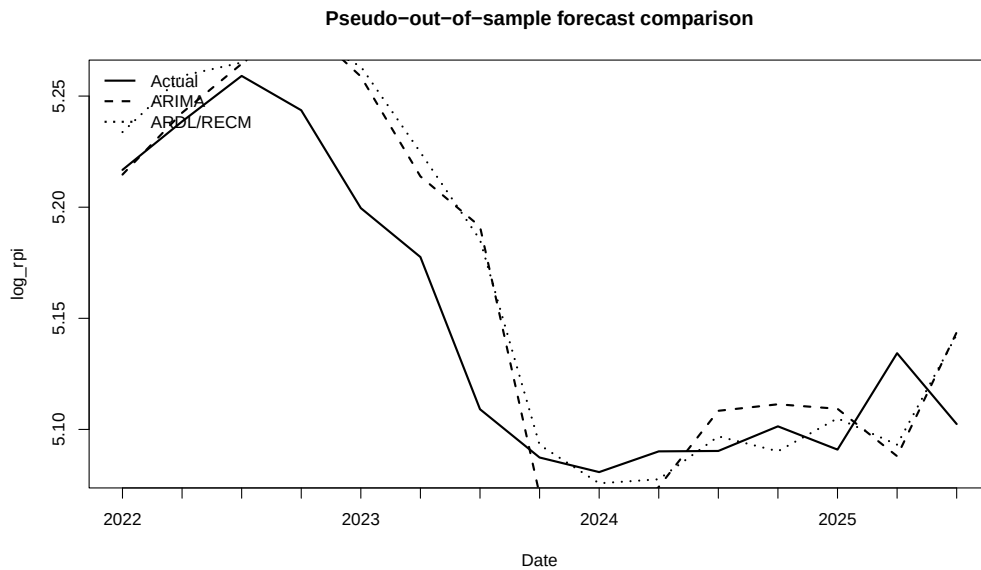


Figure 4: Pseudo out-of-sample forecast comparison between ARIMA and ARDL models.

Overall, the results suggest that while the inclusion of macroeconomic fundamentals leads to a small improvement in forecast accuracy, these gains are not statistically significant. This finding reflects the strong persistence in house prices, which is effectively captured by simple univariate time series models.

Consequently, the ARDL framework appears to offer limited advantages in short-run forecasting. Its primary contribution lies instead in providing an economically interpretable structure that links house prices to underlying macroeconomic conditions.

4.3 Local Projection Results

Figure 5 represents the impulse response of the logarithmic House Price Index (HPI) to an unexpected mortgage interest rate innovation, based on the local projection framework described above. The shaded areas represent 95% pointwise confidence intervals based on heteroskedasticity-robust (HC1) standard errors. At impact ($h = 0$), the response

of house prices is small and statistically insignificant. This suggests that housing prices do not adjust immediately to unexpected changes in mortgage interest rates, which is consistent with the well-known sluggish adjustment of housing markets.

From horizon $h = 1$ onward, the response becomes negative, although it remains statistically insignificant in the short run. A more pronounced effect emerges at medium horizons. In particular, at $h = 3$, the response is strongly negative and statistically significant at conventional levels ($\beta_3 = -0.439$, $p < 0.01$). This indicates that an unexpected increase in mortgage interest rates leads to a substantial decline in housing prices after approximately three quarters.

The negative effect persists at $h = 4$, where it remains statistically significant ($\beta_4 = -0.390$, $p = 0.031$). This suggests that the impact of interest rate shocks on housing prices is not only delayed but also sustained over several quarters.

At longer horizons ($h = 5$ and $h = 6$), the confidence intervals include zero, failing to reject the hypothesis that the impulse response is zero at the 5% significance level. It is also possible to observe that the confidence intervals widen, reflecting higher estimation uncertainty. This pattern is typically observed in local projections with limited sample sizes and increasing lag length.

Overall, the results provide clear evidence of a delayed and persistent negative response of housing prices to mortgage interest rate innovations. The absence of an immediate effect, followed by a strong decline after several quarters, is consistent with gradual adjustment mechanisms in the housing market, including search frictions, financing constraints, and delayed investment decisions.

Given the reduced-form nature of the shock, these results should be interpreted as conditional responses rather than fully structural causal effects. Nevertheless, the findings highlight the important role of financing conditions in shaping housing market dynamics in Luxembourg.

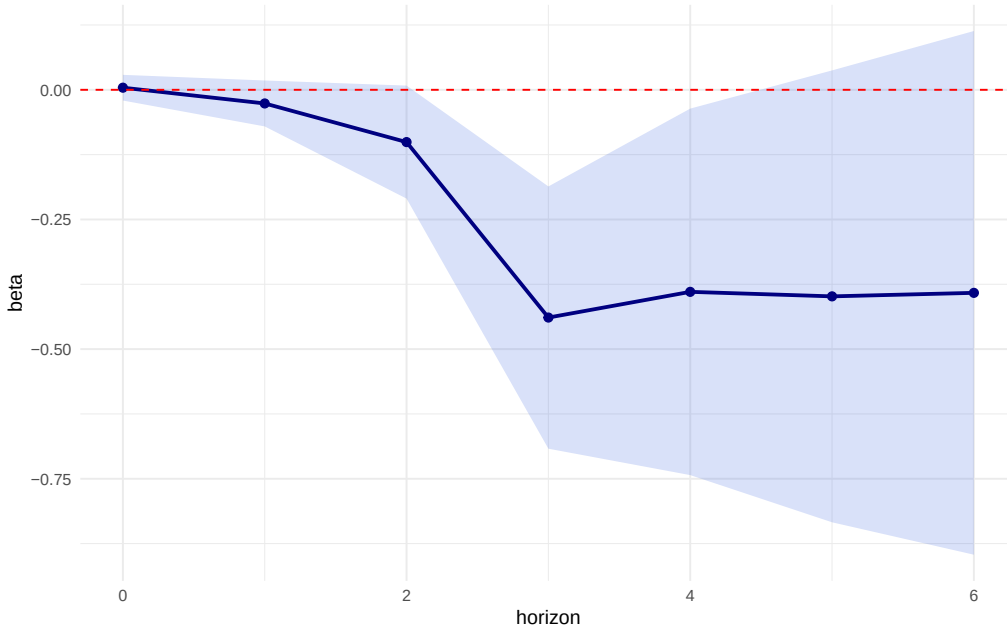


Figure 5: Impulse response of the logarithmic House Price Index (HPI) to a mortgage interest rate innovation. The solid line represents the estimated response, while the shaded area denotes 95% pointwise confidence intervals based on HC1 standard errors.

4.4 Diagnostic Test Results

Table 8 presents the results of the diagnostic tests for both the ARDL and ARIMA models.

For the ARDL model, the Breusch–Godfrey test does not reject the null hypothesis of no serial correlation ($p = 0.172$). This result is supported by the Ljung–Box test ($p = 0.772$), indicating that residual autocorrelation is not present.

The Breusch–Pagan test also fails to reject the null hypothesis of homoskedasticity ($p = 0.496$), suggesting that the variance of the residuals is stable and does not depend on the explanatory variables.

However, the Jarque–Bera test rejects the null hypothesis of normally distributed residuals ($p = 0.007$), indicating deviations from normality. While this may affect inference in very small samples, it does not compromise the consistency of the estimators. In addition, the use of heteroskedasticity-robust standard errors mitigates potential concerns regarding inference.

For the ARIMA model, the Ljung–Box test again shows no evidence of residual autocorrelation ($p = 0.681$), indicating that the model adequately captures the time series dynamics. Furthermore, the Jarque–Bera test does not reject normality ($p = 0.328$), suggesting that the residuals are approximately normally distributed.

Overall, the diagnostic results indicate that both models are well specified, with no evidence of remaining serial correlation or heteroskedasticity. The only notable issue is the non-normality of the ARDL residuals, which is common in macroeconomic time series and does not materially affect the reliability of the results.

Table 8: Diagnostic Test Results

Test	ARDL Model	ARIMA Model
Breusch–Godfrey (Serial Correlation)	0.172	–
Ljung–Box (Residual Autocorrelation)	0.772	0.681
Breusch–Pagan (Heteroskedasticity)	0.496	–
Jarque–Bera (Normality)	0.007	0.328

Note: Reported values are p-values. The null hypotheses are: no serial correlation (Breusch–Godfrey, Ljung–Box), homoskedasticity (Breusch–Pagan), and normality (Jarque–Bera).

5 Discussion

The empirical results provide several important insights into the dynamics of the Luxembourg housing market.

First, house prices exhibit a very high degree of persistence. Both the ARDL and ARIMA models indicate that current house prices are largely driven by their own past values. This strong autoregressive behaviour limits the short-run explanatory power of macroeconomic fundamentals and explains why even simple univariate models perform relatively well in forecasting exercises.

Second, the evidence for a stable long-run relationship between house prices and macroeconomic fundamentals is limited. The ARDL bounds test provides only weak or borderline evidence of cointegration, and the ECM results are not robust across specifications. While the main ECM model produces a negative and statistically significant adjustment coefficient, the robustness model does not support a stable adjustment mechanism. Therefore, the long-run equilibrium interpretation should be treated with caution.

Third, the forecasting results indicate that the inclusion of macroeconomic fundamentals leads to only modest improvements in predictive performance. While the ARDL model slightly outperforms the ARIMA benchmark in terms of RMSE and MAE, the Diebold–Mariano test shows that this difference is not statistically significant. This suggests that the strong persistence of house prices dominates short-run forecasting performance, limiting the additional predictive value of macroeconomic variables.

In addition, the slightly better out-of-sample performance of the ARDL model should be interpreted with caution. The evaluation sample includes periods of significant macroeconomic disruption, most notably the COVID-19 pandemic and the economic consequences of the Ukraine war. These events led to abrupt changes in housing market dynamics that are difficult to capture using purely autoregressive models. Since the ARIMA model relies exclusively on past values of house prices, it may struggle to adapt to such structural breaks. In contrast, the ARDL model incorporates macroeconomic variables,

which may partially reflect these disruptions and thus provide a limited improvement in forecast accuracy.

This highlights that forecast evaluation results are sensitive to the specific sample period and should be interpreted in the context of extraordinary macroeconomic conditions.

Overall, the results suggest that macroeconomic fundamentals play a more limited role in short-run forecasting than might be expected, while the dominant feature of the Luxembourg housing market remains its strong persistence. At the same time, the weak evidence for cointegration indicates that long-run equilibrium relationships are not robustly identified within the sample, further reinforcing the importance of cautious interpretation. The local projection results provide clearer evidence regarding the dynamic role of financing conditions. The impulse response function indicates that house prices do not react immediately to unexpected mortgage interest rate innovations. Instead, the response becomes negative after one quarter and reaches statistical significance at medium horizons, particularly after three to four quarters. This delayed adjustment is consistent with the slow-moving nature of housing markets, where financing conditions affect demand, borrowing constraints, and transaction decisions only gradually.

This suggests that macroeconomic fundamentals are more useful for interpreting housing market dynamics than for producing substantially better short-run forecasts. In particular, financing conditions appear to play an important role in shaping medium-term house price adjustments, even though simple autoregressive models remain difficult to outperform in short-run forecasting.

Several limitations should be acknowledged. The sample size is relatively small, which limits the precision of the estimates, especially in the local projection analysis at longer horizons. In addition, the interest-rate innovation is identified using a reduced-form residualisation approach and should therefore not be interpreted as a fully structural monetary policy shock. Finally, the results may be sensitive to model specification, as shown by the cointegration and ECM robustness checks.

6 Conclusion

This paper analysed the Luxembourg housing market using quarterly data from 2007Q1 to 2025Q3. The objective was twofold: first, to evaluate whether macroeconomic fundamentals improve house price forecasts relative to a univariate ARIMA benchmark; and second, to estimate the dynamic response of house prices to unexpected mortgage interest rate innovations.

The forecasting results show that the ARDL model slightly outperforms the ARIMA benchmark in terms of RMSE and MAE. However, the Diebold–Mariano test does not reject equal predictive accuracy. This implies that the forecasting gains from including macroeconomic fundamentals are modest and statistically insignificant. The strong per-

sistence of house prices allows the ARIMA model to perform competitively despite its purely univariate structure.

The cointegration analysis provides only weak evidence of a stable long-run relationship between house prices and macroeconomic fundamentals. While the main ECM specification suggests slow adjustment toward equilibrium, this result is not robust across specifications. Consequently, the long-run interpretation of the ARDL framework should be viewed cautiously.

The local projection results provide the strongest economic insight. An unexpected increase in mortgage interest rates is associated with a delayed and persistent decline in house prices. The response is insignificant on impact but becomes statistically significant after approximately three to four quarters. This suggests that financing conditions affect Luxembourg house prices with a delay, reflecting gradual adjustment in housing demand, credit conditions, and market activity.

In conclusion, Luxembourg house prices are highly persistent and difficult to forecast beyond their own past dynamics. Macroeconomic fundamentals offer only limited short-run forecasting gains, but they remain important for understanding the economic mechanisms behind housing market movements. In particular, the local projection results indicate that mortgage interest rate innovations play a meaningful role in medium-term house price adjustments.

Future research could extend this analysis by incorporating additional financial variables, household credit indicators, supply-side constraints, or regional housing market data. Moreover, a structural identification strategy for interest-rate shocks could provide stronger causal evidence on the transmission of monetary and financing conditions to the Luxembourg housing market.

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